

Practical 5 PKI and secure email communication

Task 1: Complete page 1 till page 16 of openssl PKI tutorial save in file pki-tutorial.pdf, stop after completing subsection 3.2 create email request.

Task 2: Use what you learn in task 1 to create an email certificate for Alice Flinstone who works in the same company. Use the following parameters to configure her certificate: DC=org, DC=simple, O=Simple Inc, CN=Fred Flintstone, emailAddress=fred@simple.org. Leave other fields empty.

Task 3: create certificates in PEM format

`openssl x509 -in certs/fred.crt -out certs/fred.pem -outform PEM`

`openssl x509 -in certs/alice.crt -out certs/alice.pem -outform PEM`

`openssl x509 -in ca/signing-ca.crt -out ca/signing-ca.pem -outform PEM`

`openssl x509 -in ca/root-ca.crt -out ca/root-ca.pem -outform PEM`

Bundle signing CA's certificate together with root CA's certificate together

`cat ca/signing-ca.pem ca/root-ca.pem >ca/signing-ca-chain.pem`

Task 3: Create a directory "emailexample" use this command `mkdir emailexample`. Go to the emailexample directory using command `cd emailexample`. Create a text file call myemail use the command `echo "how are you today?" > myemail`. You can modify the contents of myemail using any text editors such as gedit or vim.

Task 4 : Read the manual page of the command smime use command `man smime`.

Task 5 : Sign and encrypt the mail file

5.1 Email signature

```
openssl smime -sign -in myemail -text -signer ../certs/fred.pem -inkey
../certs/fred.key -from fred@simple.org -to alice@simple.org -subject
"signed email" -out signedemail
```

fred.pem is the signer's certificate and fred.key is his private key. The output has been save into the signedemail file. Use `cat signedemail` to check the content of the signed email.

5.2 Email signature verification

To verify the authenticity of the signed mail, you need to use the certificates of the CAs using the command below. This will print the content of the message and verify the validity of the certificate chain.

```
openssl smime -verify -in signedemail -CAfile ../ca/signing-ca-chain.pem
```

5.3 Email encryption

All certificates can be public, the private keys will be kept privately only known by the user himself/herself. To send an encrypted mail to a recipient, his/her certificate will be used for the encryption.

The following command shows how Fred can encrypt a message using Alice's certificate before sending the encrypted email save in file encrypted-signedemail to Alice:

```
openssl smime -encrypt -in signedemail -out encrypted-signedemail ../certs/alice.pem
```

5.4 Email decryption

Decrypt the mail using the recipient's private key. :

```
openssl smime -decrypt -in encrypted-signedemail -out received-signedemail -recip ../certs/alice.pem -inkey ../certs/alice.key
```

5.5 Email signature verification

To verify the authenticity of the signed mail, you need to use the certificates of the CAs using the following command:

```
openssl smime -verify -in received-signedemail -CAfile ../ca/signing-ca-chain.pem
```